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# A comparative study of explainable machine learning models with Shapley values for diabetes prediction

Keona Pang

*Faculty of Health Sciences, McMaster University, 1200 Main St. West, Hamilton, L8N 3Z5, ON, Canada*

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ABSTRACT

Over the years, numerous machine learning models have been developed, leading to successful applications across various fields. This study uses a large dataset related to type 2 diabetes prediction from the Centers for Disease Control and Prevention (CDC) in the United States. The dataset with 70692 samples has 21 input features and one output (non-diabetes or diabetes). In addition to health indicators like Body Mass Index (BMI), blood pressure, and cholesterol level, the features include socioeconomic factors (e.g., income, education) and lifestyle factors such as diet and physical activity. This paper aims to study how these features influence diabetes risk. 80 % of the dataset is used for training and 20 % for testing. Six machine learning models, as well as the Multivariate Adaptive Regression Splines (MARS) model, were used in the investigation. A detailed comparison of the performance of these models is given. Shapley values explain the nature of various machine learning models using visualization by color graphs to demonstrate the reliability of different machine learning models. This paper shows how Shapley values can improve their explainability and interpretability on diabetes prediction. We leverage the SHapley Additive exPlanations (SHAP) scores to provide information about the relative importance of each predictive feature, and these results shed light on the relationship between the features and the risk of developing type 2 diabetes.

## 1. Introduction

One of the advancements in AI has been the developments of various machine learning (ML) models, which have been applied across diverse domains, including clinical risk and disease prediction. The healthcare industry and hospitals generate large amounts of data on the patients over the years. These data are valuable for the discovery of unknown patterns or relationships that could be very useful for decision making. Medical data analysis for diabetes prediction using different machine learning models and a comparison of their performance results would be the focus of this paper. Explainable Artificial Intelligence (XAI) methods have appeared recently and their use for providing insights and explanations in the decision-making process of ML models are invaluable. They allow health professionals to rationalize the predicted outcomes, improve the understanding of the disease and facilitate clinical decisions.

The dataset used for this study is based on a large dataset from Centers for Disease Control and Prevention (CDC) [1]. The next section would provide a detailed discussion of the dataset and the features that are used. The features involve age, BMI, blood pressure, cholesterol level, diet, income and education of a person, who can be a diabetes or non-diabetes patient. The objective is that through the analysis of a machine learning model, one can find out the relative importance of the features involved. Some features (or information collected) from the patients could have a very low influence on diabetes prediction while some features could have a very strong (positive or negative) influence on the prediction.

Diabetes (Type 2) [2] is a chronic disease that can cause an increased risk to other diseases (e.g. heart and kidney disease, stroke, blindness), and hence early detection based on a person's features (attributes/characteristics) is important. According to World Health Organization [3], the number of people living with diabetes was 830 million in 2022. Also, 14 % of adults aged 18 years and older were living with diabetes. In addition, 59 % of adults aged 30 years and over living with diabetes were not taking medication. This is alarming and predicting the risk of developing diabetes accurately can allow for early intervention and preventive measures. This can include lifestyle changes, dietary modifications as well as early medical monitoring to delay or potentially prevent the onset of the disease.

*E-mail address:* pangk3@mcmaster.ca.

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## 2. Related works

### 2.1. Diabetes prediction using machine learning algorithms

Diabetes prediction using various machine learning algorithms is given in Ref. [4]. The dataset is the PIMA dataset [5] consisted of 768 samples with eight input features (number of times pregnant, glucose, diastolic blood pressure, triceps skin fold thickness, insulin, BMI, diabetes pedigree, age). The data is divided into training and test data with 70:30 ratio. Several machine learning algorithms are used which include support vector machine (SVM), Logistic Regression (LR) and neural network. An accuracy of around 77 % is achieved. Another study [6] that used the PIMA dataset also carry out the training with 70 % data. The K-nearest Neighbors (KNN) ML algorithm is used with an accuracy of 76 %, with precision of 80 %, recall of 85 %, F1 Score of 83 %. In Ref. [7], the PIMA dataset is also used, and SVM, Decision Tree Classifier, Random Forest, KNN, Linear regression, Logistic regression, Naive Bayes ML algorithms are used. The original dataset is not balanced, with 268 diabetic cases and 500 non-diabetic cases. The researchers used Synthetic Minority Oversampling Technique (SMOTE) to build artificial instances of the diabetes class by interpolating between existing examples. The final balanced dataset has 1000 samples, as the number of diabetic cases is also 500. The overall performance of the ML models have improved, with the Random Forest algorithm reaching an accuracy of 91.1 %.

In [8], the researchers integrated the ML model results from SVM and artificial neural network (ANN) using fuzzy logic. However, the accuracy of 66.14 % reported is not high. Another paper [9] utilizes the PIMA dataset, but the authors made use of the Long Short-Term Memory (LSTM) model of neural network for diabetes prediction. In a comparison with many other ML models, the LSTM model has achieved an impressive results of 93 % accuracy, with precision of 89 %, recall of 95 %, and F1 Score of 92 %. Yet, no details on the structure and hyper-parameters of the neural network is given in the paper. The PIMA dataset is used again in Ref. [10] for ML model and MARS model training, with MARS getting the best overall result of 78 %. Yet, there is insufficient discussion on the MARS equation obtained from the model.

In [11], another dataset called D1NAMO for diabetes prediction is used. Data from four types of sensors is used: electrocardiogram (ECG) sensor data, breathing sensor data, accelerometer (ACC) signals and glucose measurements. The multimodal dataset is small, consisting data from only 29 patients, with twenty healthy people and nine diabetes patients. The study has found that a multisensor combination using glucose, ECG, and ACC sensors gives the highest prediction accuracy of 98.2 % with the extreme gradient boosting (XGBoost) algorithm. Moreover, multisensor prediction shows nearly 4 %–5 % increase in the diabetes prediction rates as compared to single sensors.

The study in Ref. [12] used the BRFSS-2021 dataset and is focused on diabetes prediction based on the person's health and socio-economic conditions. The inputs contain 20 features, which include BMI, AGE, Income, Smoking, Blood Pressure, Cholesterol, Heart Disease, Asthma, Kidney Disease, Marital Status, Education, General Health Condition, Exercise, Arthritis, Depression, Food and Vegetable Consumption, Sex. The output is a prediction of the risk factor on diabetes. The dataset is quite large consisted of 313,913 cases. The machine learning models used for the study are logistic regression, random forest, gradient boosting and adaboosting. As the dataset is not balanced (with 48,581 diabetic cases and 265,332 non-diabetic cases), the authors have employed various sampling techniques and ensemble machine learning algorithms to address this imbalance. Data-balancing techniques have been used to improve the performance of the machine learning models, especially on the recall. The final performance of the models are in the range from 70 % to 76 %.

The dataset of the study in Ref. [13] was collected at a private medical institute in South Korea from 2013 to 2018. It has twelve features and the number of number of subjects was 253,395. Non-diabetic cases account for 68.1 % and hence the data imbalance problem is tackled by majority under-sampling and synthetic minority over-sampling (SMOTE) methods. Machine learning models of linear regression, random forest, support vector machine, XGBoost and confusion matrix-based classifier integration approach (CIM) have been used to predict the outcome. An interesting result of the paper is that if only five traditional features (FPG,HbA1c,BMI,age,sex) were used for training instead of the 12 features (FPG,HbA1c, triglycerides, BMI, gamma-GTP, age, uric acid, sex, smoking, drinking, physical activity, family history), then the maximum accuracy would be 77 % instead of 81 %. It can be concluded that the additional features contribute to improved accuracy of prediction.

In [14], a medical information system for diagnosing gestational, type 1, and type 2 diabetes using a multi-layer neural network no-prop algorithm is proposed. An individually trained multi-layer neural network is used to diagnose whether the patient falls into one of the diabetic categories. The maximum specificity and sensitivity values achieved were an impressive 95 % and 97 %, with an accuracy score of 97 %

### 2.2. Explainable AI on diabetes prediction

Explainable Artificial Intelligence (XAI) [15] is invaluable for transparency and accountability in the operations of machine learning systems. In order for a user to trust an AI system providing critical decision-making, the user needs to comprehend the fundamental justification for the system's decision and prediction. Some recent literature items on XAI for diabetes prediction is given below.

Gandhi and Mishra [16] presented various interpretable machine learning methods to help the users understand the factors related to the diabetes prediction. The methods such as Partial Dependence Plot (PDP), Individual Conditional Expectation (ICE) and Accumulated Local Effects (ALE) Plot are discussed. The LIME (Local interpretable model-agnostic explanations) method is also presented, along with the SHAP (SHapley Additive exPlanations) method. The paper presented results obtained from each method based on the Pima Indian diabetes. The paper showed that none of methods can completely be used for interpretation of outcome but the combination of various XAI models would be sufficient for giving an insight into the prediction.

Lu and Uddin [17] proposed a stacking-based model for predicting 30-day hospitalisations of diabetic patients. XAI techniques have been used to identify the most important features for prediction and increase transparency. In the study, they compared the performance of their model with other similar studies in the existing literature using the same dataset. Their model outperformed previous prediction results. The LIME (interpretable model-agnostic explanations) method was also used to demonstrate explainability at the local level for individual case.

Vishwarupe et al. [18] presented a study using a toolkit called ELI5 XAI in combination with the LIME (Local Interpretable Modelagnostic Explanations) and SHAP (Shapley Additive exPlanations) algorithmic frameworks in Python to determine whether a patient is diabetic based on a randomised clinical trial dataset. They also identified trends and the most vital factors that can help clinicians and researchers to analyze patient data and explain why the AI system has arrived at a certain decision.

In [19], this study has develop a hybrid Bayesian optimized explainable TabNet (BO-TabNet) model fpr diabetes classification. The interpretability and attention mechanism of TabNet architecture was used, which offered the local and global model explanations on the influence of the attributes on the outcomes. The model was further explained locally and globally using more robust model-agnostic LIME and SHAP XAI tools. The proposed model outperformed all benchmarked models with the use of the Pima Indians diabetes dataset (PIDD) and the Early Stage Diabetes Risk Prediction Dataset (ESDRPD).

In [20], a community local explainable agnostic model called LIME is examined for generating explanations of a chosen machine learning

model for diabetic prediction. Based on the PIMA dataset, the study stated that the community voting classifier achieved 81 % accuracy. An explainable artificial intelligence (XAI) technique is used to explain the predicted result. For example, when a test data is input, LIME would display a colored bar chart that would indicate the relative positive and negative effects of each factor towards the prediction. Such a technique would help medical professionals to understand the predictions made by the machine learning algorithm.

Uysal [21] studied the use of XAI techniques for diabetes prediction and many machine learning algorithms are examined. The SVM and random forest models are found to be most successful and hence different SHAP plots representing the contribution of each feature for the prediction are further examined. The study has helped to uncover important characteristics and patterns that contribute to diabetes risk using XAI. As expected, the factors of glucose, age and BMI are found to have significant and positive effect on the model output.

Lee et al. [22] investigated the application of the SHAP method in diabetes risk prediction models and evaluated its potential to enhance model interpretability. The results indicated that the gradient boosting decision tree excelled in diabetes risk prediction. The SHAP method provided clear and comprehensible model explanations, and helped clinicians in understanding the model's decision-making process. The SHAP method is effective in highlighting the importance of key features such as glucose, BMI, and age, which are consistent with clinically recognized diabetes risk factors. Also, it identified other significant features like blood pressure and the number of pregnancies, offering new insights for clinical decision-making. Individual case explanations provided by SHAP also illustrated how specific features influenced predictions, which would help to make personalized diagnoses and treatment plans.

In [23], a review of XAI research for diabetes prediction is given. The LIME and SHAP methods were discussed as they are prominent examples of model-agnostic methods for explaining machine learning predictions. ELI5 is a Python library that tackles the interpretability in artificial intelligence models, and provides feature importance rankings. ELI5 utilizes permutation importance, a technique that shuffles a single feature's values and observes the impact on the model's performance. The paper collected some results obtained based on two popular datasets: PIMA and ESDRPD.

Tanim et al. [24] introduced a deep neural network model called DeepNetX2 with two XAI techniques: LIME and SHAP. The study carried a data preprocessing technique that utilized Spearman's correlation coefficient. Only the most relevant features are selected for subsequent training by the deep learning network. Three datasets are used for performance evaluations and impressive results are obtained. Together with the use of XAI, a very powerful and useful strategy is developed.

In this paper, six different machine learning (ML) models plus the Multivariate Adaptive Regression Splines (MARS) [25] model have been used for diabetes prediction. The MARS model is a very powerful and less well-known ML model. The six ML models are Logistic Regression (LR) [26], Decision Trees (DT) [27], Random Forests (RF) [28], Support Vector Machines (SVM) [29], K-Nearest Neighbors (KNN) [30] and Gradient Boosting Machines (GBM) [31]. At the end of training of a machine learning model, one would wonder what and how much the model has learned from the training dataset. If one cannot explain or discuss this aspect with the patient, it is hard to win the confidence of the user. In order to demonstrate the reliability of the machine learning models, this study makes use of the Shapley values to explain the learnings of the models using visualization. This paper shows how the use of color graphs can improve their explainability and interpretability of the ML models on diabetes prediction.

## 3. Dataset for diabetes prediction and data preprocessing

The CDC dataset [1] used for this study is a balanced dataset that contains 70,692 rows, and each row contains 21 features associated with each patient, including their diabetes status (0 for non-diabetic, 1 for pre-diabetic or diabetic). These features are either self-reported by the participants through a questionnaire, or calculated variables based on individual participant response. The original dataset has 330 features (columns), but based on diabetes disease research regarding factors influencing diabetes disease and other chronic health conditions, only 21 features (with ID 0 to 20) are included in this analysis. In order to ensure the uniformity of the numerical magnitudes of the features, normalization is carried out so that the final scale of all the features is in the range between −1 and 1. Such scaling of the feature values are typical for the ease of training. The dataset is balanced with no missing values. The following table summarize the features of this study (see Table I).

## 4. Prediction based on six machine learning models and MARS model

### 4.1. Machine learning models

#### 4.1.1. Logistic regression (LR) [26]

Logistic regression is a widely used binary classification model that predicts the probability of an individual having diabetes based on various features. The sigmoid function is often used in the output.

#### 4.1.2. Decision trees (DT) [27]

Decision tree models can capture complex nonlinear relationships between features and the diabetes outcome, providing interpretable decision rules. This classifier has an internal hierarchy with nodes that indicate the characteristics of the data, and the branches would symbolize the decision rules, and final nodes show the result. In this way, it allows for a comprehensive yet simple representation of information. Here the algorithm begins at the root node and uses that data to forecast the class of the new data set. Decision tree ML model can identify important features and split the data based on these features to make predictions.

#### 4.1.3. Random forests (RF) [28]

Random Forests are an ensemble learning method that combines multiple decision trees to improve the accuracy and robustness of diabetes prediction. Nonlinear relationships can also be handled and they are less susceptible to overfitting. A Random Forest can be created by combining a number of Decision Trees. Predictions are made for each tree formed in the initial step, and the average of those results is obtained.

#### 4.1.4. Support vector machines (SVM) [29]

For classification problems, the SVM algorithm aims to create the best line or decision boundary that can segregate n-dimensional space into classes. The aim is that one can place a new data point into the correct category in a prediction. This best decision boundary is the hyperplane. SVMs can effectively handle high-dimensional data and nonlinear relationships, making them suitable for diabetes prediction tasks. SVM finds the optimal hyperplane that separates the two classes (diabetes vs. non-diabetes) with the largest margin.

#### 4.1.5. K-nearest neighbors (KNN) [30]

KNN is a non-parametric method that can identify patterns in the data and classify individuals based on their similarity to known cases of diabetes. It does not make assumptions about the underlying data distribution.

#### 4.1.6. Gradient boosting machines (GBM) [31]

Ensemble methods like Gradient Boosting can combine multiple models (e.g., decision trees) to create a strong predictive model for diabetes prediction, often achieving high performance.

**Table I**
Input features.

| ID | Name | Description of each feature | Original range of values | Scaled range for training |
|---|---|---|---|---|
| 0 | HighBP | 0 = no high BP<br>1 = high BP | [0,1] | [-1,1] |
| 1 | HighChol | 0 = no high cholesterol<br>1 = high cholesterol | [0,1] | [-1,1] |
| 2 | CholCheck | 0 = no cholesterol check in 5 years<br>1 = yes cholesterol check in 5 years | [0,1] | [-1,1] |
| 3 | BMI | Body Mass Index | [12,98] | [-1,1] |
| 4 | Smoker | Smoked over 100 cigarettes in life?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 5 | Stroke | Any stoke before?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 6 | HeartDiseaseorAttack | Any heart disease?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 7 | PhysActivity | Any physical activity in past 30 days?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 8 | Fruits | Consume fruit 1 or more times per day?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 9 | Veggies | Consume vegetables 1 or more times per day?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 10 | HvyAlcoholConsump | Heavy drinkers?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 11 | AnyHealthcare | Any health insurance?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 12 | NoDocbcCost | When you needed to see a doctor but could not because of cost?<br>0 = no 1 = yes | [0,1] | [-1,1] |
| 13 | GenHlth | Rate your general health in scale 1-5<br>1 = excellent 2 = very good 3 = good 4 = fair 5 = poor | [1,5] | [-1,1] |
| 14 | MentHlth | Days of mental health (stress, depression) not good: scale 0–30 days | [0,30] | [-1,1] |
| 15 | PhysHlth | Days of physical health not good: scale 0–30 days. | [0,30] | [-1,1] |
| 16 | DiffWalk | Any serious difficulty walking or climbing stairs? 0 = no 1 = yes | [0,1] | [-1,1] |
| 17 | Sex | 0 = female 1 = male | [0,1] | [-1,1] |
| 18 | Age | 13-Level age category<br>1 = 18–24 9 = 60–64<br>13 = 80 or older | [1,13] | [-1,1] |
| 19 | Education | Scale 1–6; 1 = never attended school, 6 = college graduate | [1,6] | [-1,1] |
| 20 | Income | Scale 1–8; 1 = less than $10 K; 8 = $75 K or more | [1-8] | [-1,1] |

### 4.2. MARS models

The above machine learning models are commonly used machine learning algorithms. In this study, they have all been implemented for classification tasks on the CDC Diabetes dataset, with the use of 80 % for training and 20 % for testing. A comparison of their classification results is given along with the results based on another important model, the Multivariate Adaptive Regression Splines (MARS) model.

Multivariate Adaptive Regression Splines (MARS) is a form of regression analysis method introduced by Jerome H. Friedman in 1991 [25]. It is a non-parametric regression technique and can be seen as an extension of linear regression models that automatically represents nonlinearities and interactions between variables. It has been found to be very successful in many applications [32–34]. As a machine learning model, the main advantage is that the output is an explicit equation consisted of some weighted sum of the input features and their linear or nonlinear interactions. The function equation can be readily understood, which means that the machine learning model is completely explainable. The prediction can be easily interpreted. To the user, the MARS machine learning model is not a "black box" but an explicit equation (can be linear or nonlinear) relating the output (predicted result) and the input features. After learning from the dataset, the output has the general form as follows:

$$y = \beta_0 + \sum_{k=1}^{M} \beta_k F_k(x)$$

where y is the output, x defines the input features, $\beta_0$ gives the intercept coefficient, $\beta_k$ gives the coefficient of the $k^{th}$ basis function, M is the total number of basis functions. For degree 1 MARS model, the basis functions are linear terms. For degree 2 model, the basis functions would be quadratic.

Based on the CDC Diabetes, again with a 80 % training, 20 % testing, the classification results of the test set is summarized along with the previous ML models (see Table 2).

## 5. MARS results and explanations

### 5.1. Results of the MARS degree 1 model

For a degree one model, the MARS model equation is given below:

y = 1.1195 + 0.2138*GenHlth +0.1555*Age +0.0554*HighChol +0.0787*HighBP +0.0827*CholCheck +0.027*HeartDiseaseorAttack +0.0219*Sex - 0.5319*max(0, 0.8605 - BMI), - 0.01830* PhysHlth - 0.0642* HvyAlcoholConsump - 6.8341*max(0, BMI - 0.8605) - 0.0429*Income.

One excellent feature of the machine learning model obtained by MARS is that the predicted output is an explicit function of the input feature values. For a degree 1 model, as the degree of each basis function is constrained to one, the equation would be like a linear weighted combination of the input feature, except that there could be hinge functions involved that provide some nonlinear relationships between the terms. The degree 2 model, the equations would be more complicated because the basis functions would be linear terms (like degree 1 model) plus second order terms or product of two input features, as well as the hinge functions. The equation of the degree 2 model will be shown in the next section.

The feature importance based on the MARS degree 1 model is shown in Fig. 1 below:

The MARS degree 1 model has identified only 9 features out of the 21 input feature. The top five features are "General Health","HighBP", "age", "BMI" and "HighChol".

Example (non-diabetes): The following is the row of input feature scaled values of a person [x0, …,x20].

[-1, 1, 1, −0.767, −1, −1, −1, 1, −1, 1,
−1, 1, −1, −1, −1, −1, −1, 1, −0.333, 1, 1]

The output from the degree 1 model equation gives **0.0818** or a low value in a scale [0,1], indicating the person is non-diabetes.

Example (diabetes): The following is the row of input feature (scaled values) of a person [x0, …,x20].

[ 1, 1, 1, −0.465, −1, −1, −1, 1, −1, 1,
−1, 1, −1, −0.5, −0.333, −1, −1, −1, 0.333, 0.6, −0.714 ]

**Table 2**
Classification results

| Models | Accuracy | Precision | Sensitivity-Recall | Specificity | F1 score | AUC-ROC score |
|---|---|---|---|---|---|---|
| Logistic regression | 74.7 % | 73.6 % | 76.3 % | 73.2 % | 0.749 | 0.747 |
| Decision trees | 65.8 % | 65.6 % | 64.9 % | 66.6 % | 0.653 | 0.658 |
| Random forests | 74.4 % | 72.1 % | 78.8 % | 70 % | 0.753 | 0.744 |
| Support vector machines | 74.4 % | 71.8 % | 80.5 % | 68.9 % | 0.759 | 0.747 |
| K-nearest neighbors | 68 % | 71.4 % | 58.8 % | 76.9 % | 0.645 | 0.679 |
| Gradient boosting machines | 74.9 % | 72.5 % | 79.4 % | 71.5 % | 0.758 | 0.747 |
| MARS deg 1 | 74.8 % | 73.2 % | 77.5 % | 72.1 % | 0.753 | 0.748 |
| MARS deg 2 | 74.8 % | 72.9 % | 78.2 % | 71.5 % | 0.755 | 0.748 |

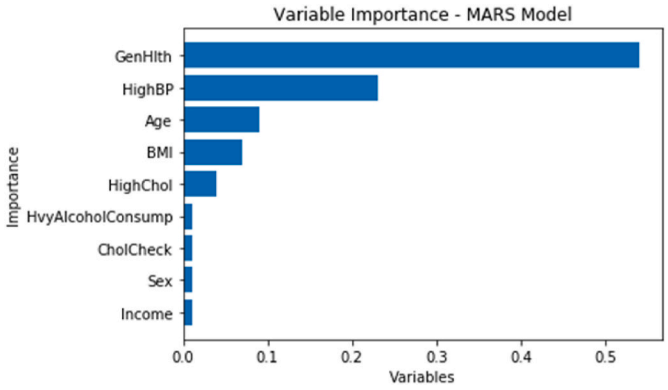



**Fig. 1.** Feature importance from MARS degree 1 model.

The output from the degree 1 model gives **0.64** or a high value in a scale [0,1], indicating the person is diabetes.

Below is the detail of the calculation involved.

| | Basis Function | Scaled value (SV) | Coefficient (Coef) | SV*Coef |
|---|---|---|---|---|
| 1 | (Intercept) | | 1.1195 | 1.1195 |
| 2 | GenHlth | −0.5 | 0.2138 | −0.1069 |
| 3 | HighBP | 1 | 0.0787 | 0.0787 |
| 4 | max(0,BMI-0.8605) | BMI = −0.465; max(0, BMI-0.8605) = 0 | −6.8341 | 0 |
| 5 | max(0,0.8605-BMI) | BMI = −0.465; max (0,0.8605-BMI) = 1.3255 | −0.5319 | −0.7070 |
| 6 | Age | 0.333 | 0.1555 | 0.0518 |
| 7 | HighChol | 1 | 0.0554 | 0.0554 |
| 8 | HvyAlcohol Consump | −1 | −0.0642 | 0.0642 |
| 9 | CholCheck | 1 | 0.0827 | 0.0827 |
| 10 | HeartDisea seorAttack | −1 | 0.0270 | −0.0270 |
| 11 | Income | −0.714 | −0.0429 | 0.0306 |
| 12 | Sex | −1 | 0.0219 | −0.0219 |
| 13 | PhysHlth | −1 | −0.0183 | 0.0183 |

Column sum = 0.64 (the output value).

### 5.2. Discussion of degree 1 model

1. With reference to row 2, the current scaled value of GenHlth is **-0.5** (i.e. very good, or value of 2 in the original scale). What if the GenHlth scaled value becomes **0.5** (i.e. fair, or value of 4 in the original scale)? The calculation in the second row would be changed to 0.5*0.2138 = 0.1069, or the overall sum would have jumped to 0.8538. The MARS equation helps us to understand better how a prediction can be made, and answer "what if" questions. Fig. 2 below is the output for the five scaled value of −1, −0.5, 0, 0.5 and 1, which are associated with the person's perception of health in the scale 1 (excellent), 2 (very good), 3(good), 4(fair) and 5(poor) respectively.
2. What if the person is a heavy drinker? The current scaled value (row 8) is −1 indicating that the person is not a heavy drinker. If that scaled value is changed to 1 (drinker), the new output value would be 0.5116. The risk of being diabetes has actually reduced. Looking at the original drinking definition in the survey, the survey question was referring to moderate alcohol consumption (up to 1 drink per day for women and up to 2 drinks per day for men) rather than heavy alcohol consumption (more than 3–4 drinks per day). Studies [35, 36] have indeed shown that moderate alcohol intake is associated with a lower risk of developing type 2 diabetes compared to abstaining from alcohol or heavy drinking. This is because the potential mechanisms include improved insulin sensitivity and reduced inflammation.
3. The output result (0.64) above is based on the person having an Age of scaled value 0.333 (row 6; actual age is 60–64). What if the actual age were 35–39 (or a scaled value of −0.5)? The output value would drop to 0.51 and will be at the borderline of deciding between

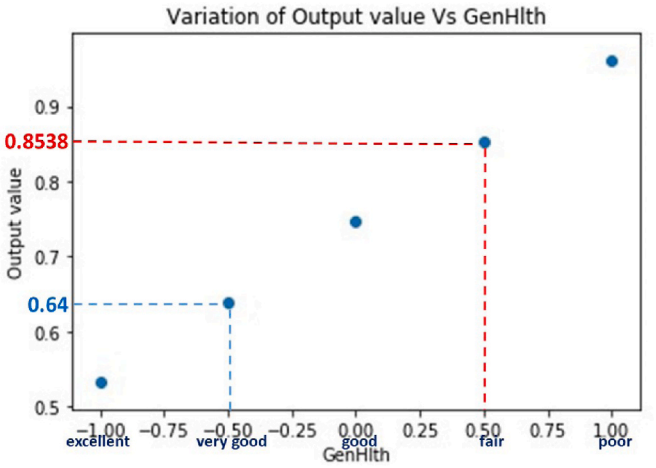



**Fig. 2.** Output for different value of GenHlth.

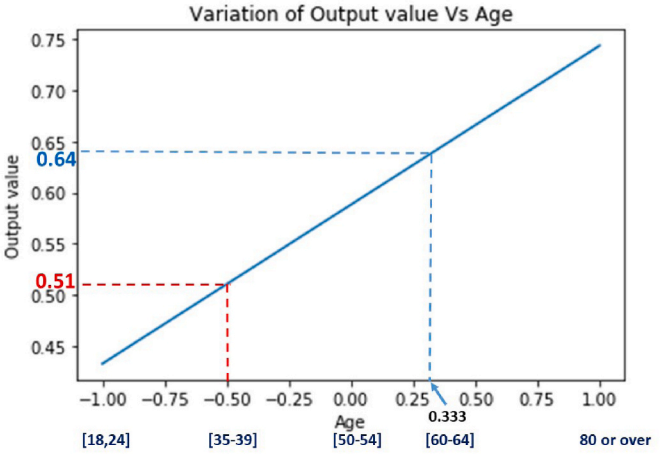



**Fig. 3.** Output for different value of age.

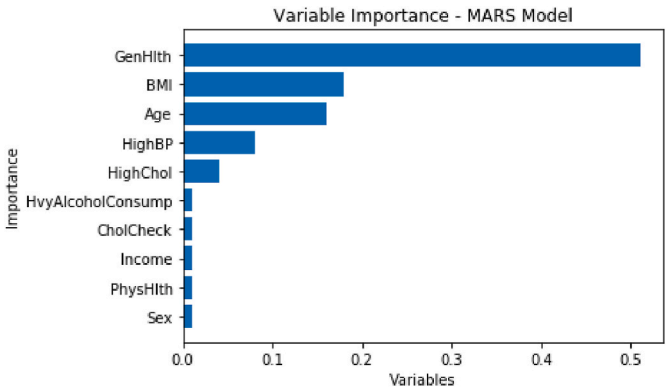



**Fig. 4.** Feature importance from MARS degree 2 model.

diabetic and non-diabetic. The variation of the output with Age is shown by the linear graph Fig. 3 below:

Older adults are known to be more vulnerable to chronic diseases [37] and its age is known to be an independent risk factor for cardiovascular diseases [38]. Although no study has shown a direct relationship between the age of a person and the risk of diabetes, study [39] has shown that a significantly higher proportion of patients ≥60 years suffered from reported diabetes-related complications. Based on the MARS degree 1 machine learning model, there is a linear relationship between age and increased risk of type 2 diabetes.

4. The current BMI has an actual value of 35 with a scaled value of −0.465.

What if actual BMI = 55 (scaled is 0), the output will become 0.89.

What if actual BMI = 20.6 (scaled value of −0.8), the output will be 0.46 or may be even classified as non-diabetes. In the medical literature, the relationship between BMI and diabetes prediction is well-established [40,41]. Higher BMI is strongly associated with an increased risk of developing type 2 diabetes. The mechanism is believed to be that excess body fat, especially abdominal/visceral fat, can lead to insulin resistance, which is a precursor to type 2 diabetes. This insulin resistance makes it harder for the body to effectively use insulin to regulate blood sugar levels.

The above analysis based on the MARS° 1 equation gives us a lot of insights and explanations on the machine learning predictions, which cannot be provided by other machine learning models.

### 5.3. Results of the MARS degree 2 model

In the next section, the function equation obtained from MARS degree 2 model is given and discussed.

The equation is:

y = 0.1468 + 0.0795*CholCheck +0.4941*max(0, 0.8605 - BMI) + 0.0634 *HighChol + 0.0742*HighBP + 0.2998*Age + 0.1742*GenHlth - 0.0616* HvyAlcoholConsump +
0.2471*Age*BMI +
0.5425*BMI*max(0, 0.8605 - BMI) +
0.0414*Sex*Age +
0.0212*HeartDiseaseorAttack*max(0, 0.8605 - BMI) +
−0.0597*GenHlth*PhysHlth.
−0.0311*Income*max(0, 0.8605 - BMI).
−0.0333*HighChol*Age.

The feature importance based on the MARS degree 2 model is shown in Fig. 4 below:

The MARS degree 2 model has identified 10 features out of the 21 input feature. The top five features are identical to the model 1 features. One minor feature "PhysHlth" is added to the list of features. The top five features ("General Health","HighBP", "age", "BMI" and "HighChol") all appear as linear terms in the equation. Some second order terms appear like "Age*BMI" or "BMI*max(0, 0.8605 - BMI)". The function equation is a lot more complex than the degree 1 equation. Yet, it still gives us the clear explicit relationship between the output (risk prediction of diabetes) and the input features.

Example (non-diabetes): The following is the row of input feature scaled values of a person [x0, …,x20].

[-1, 1, 1, −0.767, −1, −1, −1, 1, −1, 1,
−1, 1, −1, −1, −1, −1, −1, 1, −0.333, 1, 1 ]

The output from the MARS degree 2 model equation gives 0.0454 or a low value in a scale [0,1], indicating the person is non-diabetes.

Below is the detail of the calculation involved.

| | Basis Function | Scaled value (SV) | Coefficient (Coef) | SV*Coef |
|---|---|---|---|---|
| 1 | (Intercept) | | 0.1468 | 0.1468 |
| 2 | GenHlth | −1 | 0.1742 | −0.1742 |
| 3 | HighBP | −1 | 0.0742 | −0.0742 |
| 4 | Age | −0.333 | 0.2998 | −0.0998 |
| 5 | max(0,0.8605-BMI) | BMI = −0.767; max(0,0.8605-BMI) = 1.6275 | 0.4941 | **0.8042** |
| 6 | HighChol | 1 | 0.0634 | 0.0634 |
| 7 | CholCheck | 1 | 0.0795 | 0.0795 |
| 8 | HvyAlcoholConsump | −1 | −0.0616 | 0.0616 |
| 9 | BMI*max(0, 0.8605 - BMI) | BMI = −0.767 max(0, 0.8605 - BMI) = 1.6275; product gives −1.2483 | 0.5425 | **−0.6772** |
| 10 | Sex*Age | Sex = 1; age = −0.333; product gives −0.333 | 0.0414 | −0.0138 |
| 11 | HeartDiseaseorAttack*max(0, 0.8605 - BMI) | HeartDiseaseorAttack = −1; max(0, 0.8605 - BMI) = 1.3255; product = −1.6275 | 0.0212 | −0.0345 |
| 12 | Age*BMI | Age = −0.333; BMI = −0.767; product = 0.2554 | 0.2471 | 0.0631 |
| 13 | GenHlth*PhysHlth | GenHlth = −1; PhysHlth = −1; product = 1 | −0.0597 | −0.0597 |
| 14 | Income*max(0, 0.8605 - BMI) | Income = 1; max(0, 0.8605 - BMI) = 1.3255; product = 1.6275 | −0.0311 | −0.0506 |
| 15 | HighChol*Age | HighChol = 1; Age = −0.333; product = −0.333 | −0.0333 | 0.0111 |

Column sum = 0.0454 (the output value).

### 5.4. Discussion of the MARS degree 2 model

1. With reference to row 2, the current scaled value of GenHlth is −1 (i.e. excellent, or value of 1 in the original scale). What if the GenHlth scaled value becomes 1 (i.e.poor, or value of 5 in the original scale)? The calculation in the second row would be changed to 0.1742. In addition, the calculation of the second order term "GenHlth*PhysHlth) would change to 0.0597. The overall sum would increase to 0.5132. The MARS equation helps us to understand better how a prediction can be made, and answer "what if" questions. Fig. 5

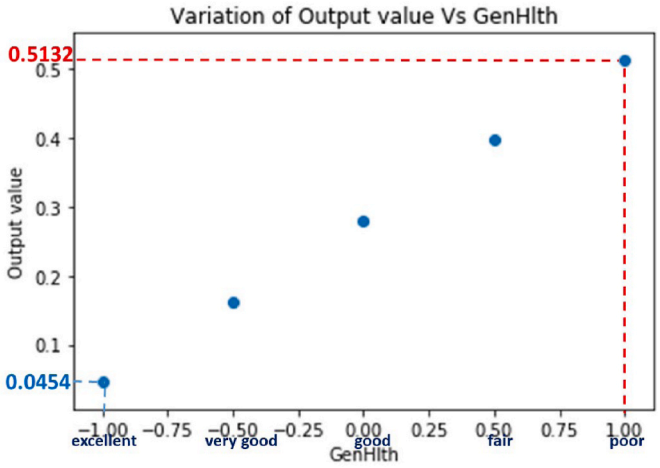



**Fig. 5.** Output for different value of GenHlth.

is the output for the five scaled value of −1, −0.5, 0, 0.5 and 1, which are associated with the person's perception of health in the scale 1 (excellent), 2 (very good), 3(good), 4(fair) and 5(poor) respectively.

2. The output result (0.0454) above is based on the person having an Age of scaled value −0.333 (row 6; actual age is 40–44). What if the actual age were 60–64 (or a scaled value of 0.333)? Row 4 would be 0.0998 instead of −0.0998, or a jump of around 0.2. Rows 10, 12 and 15 would be changed as well. The output value would increase to 0.1242. The variation of the output with Age is shown by Fig. 6 below:

   For a continuous variation of the Age value from −1 to 1.

3. It has been described that increased BMI is a robust risk factor for type 2 diabetes [40].

In this case, BMI = −0.767 (actual BMI is 22) output = 0.0454. With an increased value of BMI (actual value of 55, with a scaled value of 0), the output is increased to 0.321.

4. As discussed before, what if the person is a (moderate) drinker? The current scaled value (row 6) is −1 indicating that the person is not a drinker. The associated output is 0.0454. If that scaled value is changed to 1 (drinker), the new output value would be −0.0778, or a reduction of risk.

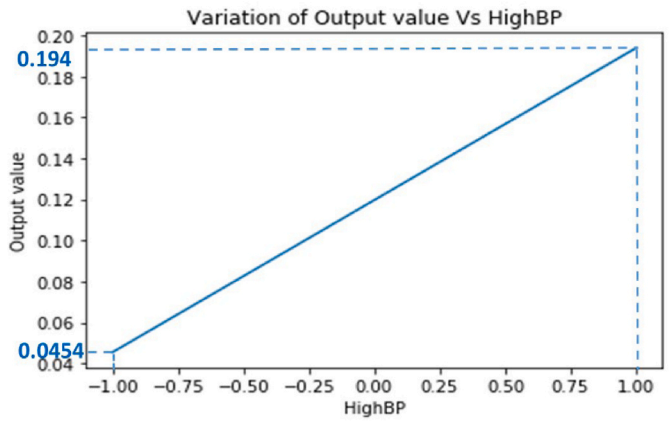



**Fig. 7.** Output for different value of HighBP.

5. Regarding the input feature of HighBP, the current scaled value is −1, which is associated with "no High Blood Pressure". As we have the equation that relate the inputs with the output, suppose "HighBP" is varied in the range [-1,1], the following result (Fig. 7) is obtained:

HighBP with scaled values of −1 and 1 are associated with output values of 0.0454 and 0.194 respectively. It must be mentioned that the above result is only a kind of "interpolation" using the scaled value of −1 (for no High Pressure" and scaled value of 1 (for High Pressure). The association of high pressure with increased risk of diabetes is very clear from the graph.

6. Regarding the input feature of "HighChol", the current scaled value is 1, which is associated with "High Cholesterol". As we have the equation that relate the inputs with the output, suppose "HighChol" is varied in the range [-1,1], the following result (Fig. 8) is obtained:

HighChol with scaled values of −1 and 1 are associated with output values of −0.104 and 0.0454 respectively. It must be mentioned that the above result is only a kind of "interpolation" using the scaled value of −1 (for no High Cholesterol" and scaled value of 1 (for High Cholesterol). The association of high cholesterol with increased risk of diabetes is also very clear from the graph.

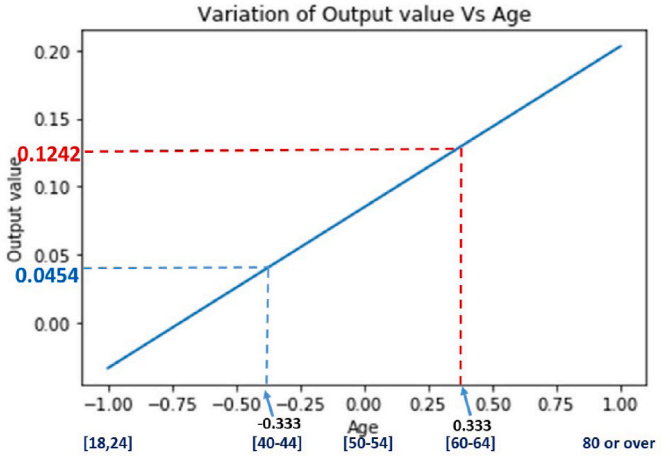



**Fig. 6.** Output for different value of age.

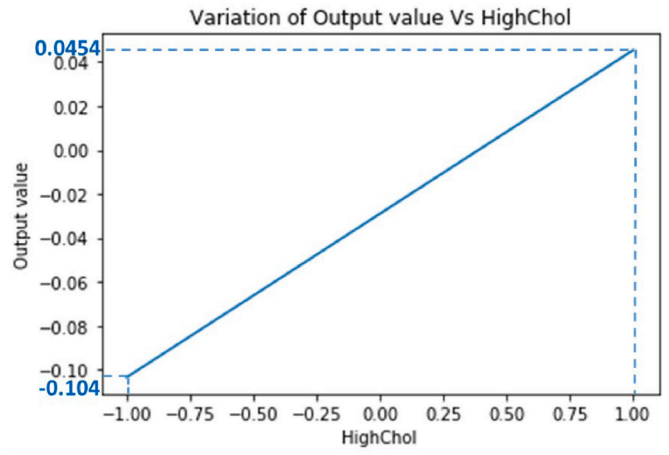



**Fig. 8.** Output for different value of HighChol.

## 6. Explanation capability based on the Shapley values

### 6.1. SHAP (Shapley Additive exPlanations) values

SHAP (Shapley Additive exPlanations) values [42,43] are a powerful method to explain the output of any machine learning model. They are based on cooperative game theory and provide a fair approach of distributing "credit" for a prediction among the different features. Each feature contributes something to the model's prediction and SHAP values tell you exactly how much credit each feature deserves for that prediction. There are two aspects on their usage. SHAP values can tell you which features are important across your entire dataset, and hence a global view of feature importance. On the other hand, SHAP values tell you how a feature impacts each individual prediction. Hence, it provides a local view and helps to user to understand why the model makes a specific prediction.

After training a machine learning model based on the dataset, we can calculate the SHAP values for each prediction using the SHAP library. SHAP values tell you how each feature contributed to the prediction for each case. SHAP offers various plots for interpreting the results.

- Feature Importance Bar Chart: It gives a global view of feature importance across all predictions based on the entire dataset.
- SHAP Summary Plot: This plot shows the impact of all features across all predictions. Each point represents a SHAP value for a feature of an instance. The color indicates the feature value. If a feature has a wide spread in the SHAP summary plot, it means its impact varies significantly across instances, and thus sometimes increasing and sometimes decreasing the prediction.
- SHAP Waterfall Plot: It is for explaining individual prediction. It gives the details of one prediction by showing how each feature pushed the prediction up or down from the base value. One can use this plot as justifications why the model makes a certain decision.

Overall, the visualizations give granular insights and show feature importance and interpretation for individual predictions. Next, the SHAP library in python is used to provide the interpretability for the MARS degree 1 model. The feature importance bar chart is shown in Fig. 9 below:

Note that the above chart is very similar to the feature importance information obtained by MARS degree 2 model in Fig. 4. It is interesting to note that the top feature is GenHlth, which is an integer value from 1 (excellent health) to 5 (poor health), reported by an individual based on their own health perception. Yet, the above does not show whether the influence is positive or negative. In a scaled value of GenHlth, a value of $-1$ means excellent health (negative influence on being diabetes) whereas $+1$ means poor health (positive influence on being diabetes).

Fig. 10 shows the SHAP Summary Plot of the dataset. Each point represents a row of data from the dataset.

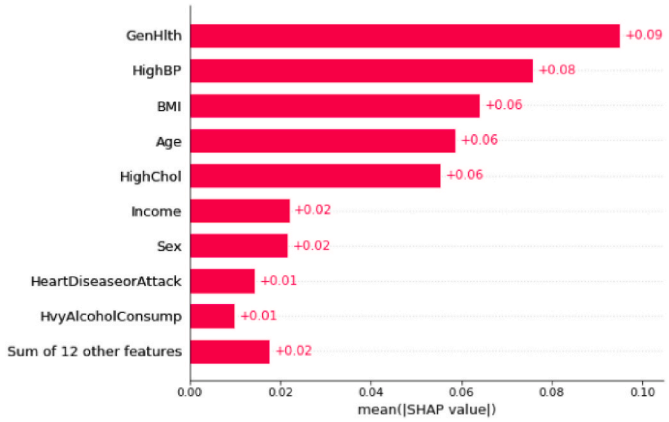



**Fig. 9.** Feature importance bar chart obtained from SHAP.

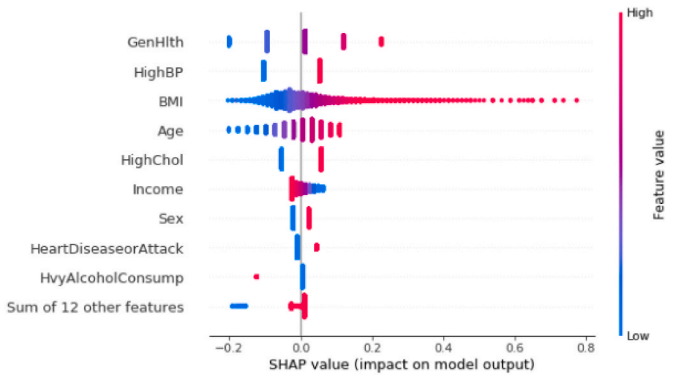



**Fig. 10.** SHAP summary plot based on the dataset.

### 6.2. Discussion

1. It shows clearly that high values (shown in red) of the GenHlth, HighBP, BMI, Age, HighChol etc. have a positive influence/effect on being diabetes. These are shown as red dots on the right hand side of the SHAP scale of the x-axis. Note that the feature BMI has a wide spread in the plot, which means its impact varies significantly across instances.
2. Note that Income (6th item) has red dotes on the left hand side, which means that high values of Income has a negative influence on being diabetes. Individuals with higher income typically have better access to healthcare resources, including preventive services. This can lead to earlier detection and more consistent management of diabetes, potentially reducing complications. Indeed, research has generally shown that lower income levels are often associated with higher rates of diabetes [44].
3. The seventh item is Sex, and it is interesting to note the red dots (indicating high values of $+1$ or male) are on the right hand side. It means that being male has a positive influence on being diabetes. Research has also found that men are at a higher risk of developing type 2 diabetes compared to women [45]. For men, testosterone levels have been linked to insulin sensitivity, with lower testosterone levels often associated with a higher risk of developing type 2 diabetes. Men may also have less frequent doctor visits and poorer adherence to diabetes management plans when compared to women, potentially leading to poorer outcomes.

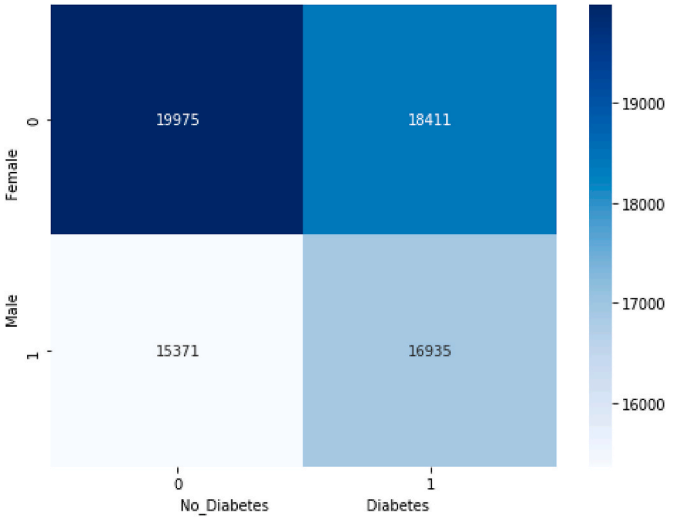



**Fig. 11.** Number of diabetes and non-diabetes for male and female.

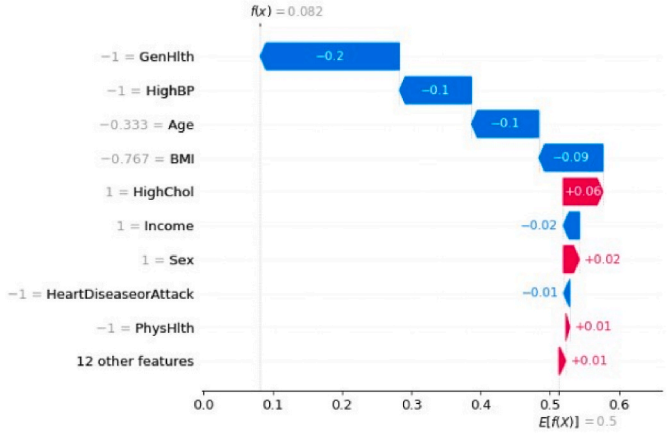



**Fig. 12.** Analysis based on the features of a non-diabetes.

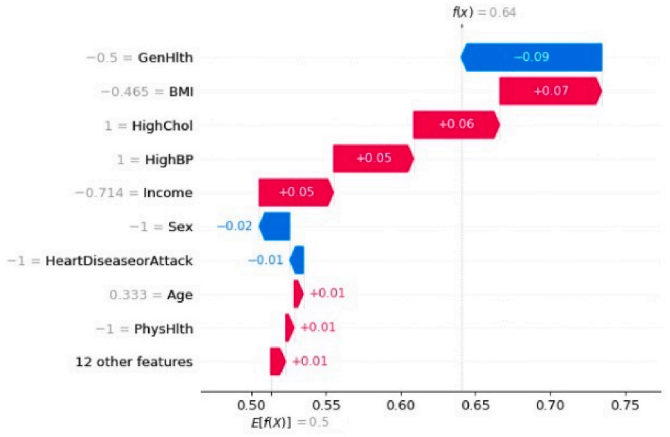



**Fig. 13.** Analysis based on the features of a diabetes.

The observation can also be found from a careful analysis on the original dataset. Fig. 11 shows the distribution of non-diabetes and diabetes for male and female. Of the 32,306 males, there are more diabetes (16,935) than diabetes. However, of the 38,386 females, there are more non-diabetes (19,975).

The Shapley values can also be used to indicate why an input is giving a certain prediction from the model.

### 6.3. *Example 1 (non-diabetes): the following is the original row of input feature values of a person*

[-1, 1, 1, −0.767, −1, −1, −1, 1, −1, 1,
−1, 1, −1, −1, −1, −1, −1, 1, −0.333, 1, 1 ]

The output from the model gives 0.082 which is a low value in a scale [0,1], indicating the person is non-diabetes. The details on how this value is obtained will be explained in this section.

When provided with the input feature values to the model, Fig. 12 gives the Waterfall Plot from SHAD, which explains how the output of 0.082 is obtained. The SHAD values show how each feature affects the final prediction, as well as the significance of each feature compared to others. These values are shown in red or blue arrows depending on whether they increase or decrease the predicted value in the SHAP Waterfall Plot of Fig. 12 below. It shows how the SHAP values move the model's prediction from the expected value *E[f(x)]* 0.5 displayed at the bottom of the of the chart to the predicted value *f(x)* 0.082 at the top. The magnitude of the arrows is a measure of the strength of the feature. The original data on the feature values are shown on the left-hand-side of the picture.

1. The features near the top are the more important features contributing to the influence on diabetes. As discussed, General health is at the top. The person has a scaled value of −1 (meaning he feels excellent health in the scale [-1,1]), and the associated SHAP value is −0.2, indicating a very negative influence on being diabetes. All the features with negative influence (in blue) are summarized below:

| Feature | Scaled value | Meaning | Shad value |
|---|---|---|---|
| GenHlth (general Health) | −1 | Excellent health | −0.2 |
| HighBP | −1 | No high blood pressure | −0.1 |
| Age | −0.333 | Age 40-44 | −0.1 |
| BMI | −0.767 | Actual BMI around 22 | −0.09 |
| Income | +1 | High income | −0.02 |
| HeartDisease or attack | −1 | No heart disease or attack | −0.01 |

Total = −0.52.

2. Features with positive influence (in red) of being diabetes:

| Feature | Scaled value | Meaning | Shad value |
|---|---|---|---|
| HighChol | +1 | High cholesterol | +0.06 |
| Sex | +1 | Male | +0.02 |
| PhysHlth | −1 | Physical health | +0.01 |
| Other features | +1 | | +0.01 |

Total = +0.10.

From above, the sub-total of the Shad values from features with negative influence is −0.52. The sub-total of the Shad values from features with positive influence is 0.1. Hence, the overall SHAD value = 0.5–0.52 + 0.1 = 0.08 (predicting a non-diabetes). The output of 0.082 is just additional degree of accuracy given by the software. Note that the base value is 0.5.

### 6.4. *Example 2 (diabetes): the following is the original row of input feature values of a person*

[ 1, 1, 1, −0.465, −1, −1, −1, 1, −1, 1,
−1, 1, −1, −0.5, −0.333, −1, −1, −1, 0.333, 0.6, −0.714 ]

The output from the model gives 0.64 or a high value in a scale [0,1], indicating the person is diabetes.

When provided with the input feature values to the model, the following SHAP Waterfall Plot of Fig. 13 explains how the output of 0.64 is obtained.

1. Features with negative influence being diabetes are summarized below:

| Feature | Scaled value | Meaning | Shad value |
|---|---|---|---|
| GenHlth (general Health) | −0.5 | Very good | −0.09 |
| Sex | −1 | Female | −0.02 |
| HeartDisease or attack | −1 | No heart disease or attack | −0.01 |

Total = −0.12.

2. Features with positive influence of being diabetes:

| Feature | Scaled value | Meaning | Shad value |
|---|---|---|---|
| BMI | −0.465 | Actual BMI is 35, a bit high | +0.07 |
| HighChol | +1 | High cholesterol | +0.06 |
| HighBP | +1 | High blood pressure | +0.05 |
| Income | −0.714 | Low income | +0.05 |
| Age | 0.333 | Age 60-64 | +0.01 |
| PhysHlth | −1 | Physical health | +0.01 |
| Other features | +1 | | +0.01 |

Total = +0.26.

From above, the sub-total of the Shad values from features with negative influence is −0.12. The sub-total of the Shad values from features with positive influence is 0.26. Hence, the overall SHAD value = 0.5–0.12 + 0.26 = 0.64 (predicting a diabetes).

## 7. Conclusions

In this paper, a large CDC dataset relating 21 features and type 2 diabetes is used for developing machine learning models for diabetes prediction. Six traditional machine models were developed in addition to models based on Multivariate Adaptive Regression Splines (MARS). The MARS model can provide explicit equation relating functional relationship between the input features and the output diabetes prediction. Such function can provide good explanations to the user, and answer "what-if" questions. Detailed analysis and discussion of the diabetes database and features are given in the paper.

Another contribution of this paper is that SHAP values have been used to provide the explainability and interpretability of the MARS machine learning model on diabetes prediction. While there are many machine learning models utilized by researchers across many fields, the accuracy and transparency of a model is particularly important in the context of healthcare-related prediction. It is highly desirable to utilize a model Tat can explain why a certain prediction is made. It should be able to justify the model's prediction and answer "what-if" kind of questions. In this paper, with many detailed examples, it has been shown how the SHAP values can help to explain individual predictions. The essential features that have lead that prediction is explained, which can help the users to understand and trust a model's prediction.

This study has limitations like many clinical prediction efforts. The work needs to be tested and validated on more datasets. We will explore other machine learning or deep learning methods like neural networks to increase prediction performance in the future. Other XAI methods like LIME, Partial Dependence Plot (PDP), Individual Conditional Expectation (ICE) and Accumulated Local Effects (ALE) Plot will also be examined.

## Declaration of competing interest

I would like to declare that I have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

I do not have any financial and personal relationships with other people or organizations that could inappropriately influence (bias) my work.

I have no interests to declare.

## Data availability

Data will be made available on request.

## References


[1] CDC, Diabetes dataset, diabetes_binary_5050split_health_indicators_BRFSS2015.csv, Accessed April 2, https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset, 2024.

[2] CDC, Type 2 diabetes (2024). Accessed June 15, https://www.cdc.gov/diabetes/about/about-type-2-diabetes.html.

[3] World Health Organization, Diabetes, World Health Organization, December 15, 2024 [Online], https://www.who.int/news-room/fact-sheets/detail/diabetes.

[4] K.V. Haranadh, K. Chakri Dhanush, Prediction of diabetes using machine learning algorithms, International Journal of Scientific Research in Engineering and Management 8 (4) (April 2024).

[5] PIMA database. https://data.world/data-society/pima-indians-diabetes-database. (Accessed 15 March 2024).

[6] S. Mansouri, S. Boulares, S. Chabchoub, Machine learning for early diabetes detection and diagnosis, Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications 15 (1) (March 2024) 216–230.

[7] G. Saranya, S.D. Pande, Enhancing diabetes prediction with data preprocessing and various machine learning algorithms, EAI Endorsed Transactions on Internet of Things 10 (March 2024).

[8] S.R. Kumar, G. Kruthi, V. Supraja, Diabetes prediction with machine learning with Python", International Journal of Scientific Research in Computer Science, Engineering and Information Technology (2024) https://doi.org/10.32628/IJSRCSEIT.

[9] N. Tripathy, B. Moharana, S.K. Balabantaray, S.K. Nayak, A. Pati, A. Panigrahi, A comparative analysis of diabetes prediction using machine learning and deep learning algorithms in healthcare, in: International Conference on Advancements in Smart, Secure and Intelligent Computing (ASSIC), January 2024.

[10] D. Senthilkumar, S. Paulraj, Diabetes disease diagnosis using multivariate adaptive regression splines, Int. J. Eng. Technol. 5 (2013), 5.

[11] A. Site, J. Nurmi, E.S. Lohan, Machine-learning-based diabetes prediction using multi-sensor data, IEEE Sens. J. 23 (22) (October 2023).

[12] M.M. Chowdhury, R.S. Ayon, M.S. Hossain, An investigation of machine learning algorithms and data augmentation techniques for diabetes diagnosis using class imbalanced BRFSS dataset, Healthcare Analytics 5 (2024).

[13] H.M. Deberneh, I. Kim, Prediction of type 2 diabetes based on machine learning algorithm, Int. J. Environ. Res. Public health 18 (6) (2021) 3317.

[14] J.J. Sonia, P. Jayachandran, A. Quadir, S. Mohan, A.K. Sivaraman, K.F. Tee, Machine-learning-based diabetes mellitus risk prediction using multi-layer neural network No-prop algorithm, Diagnostics 13 (2023) 723, https://doi.org/10.3390/diagnostics13040723.

[15] IBM, What is explainable AI?, Armonk, New York, United States [Online]. Available: https://www.ibm.com/topics/explainable-ai. (Accessed 13 December 2024).

[16] N. Gandhi, S. Mishra, Explainable AI for healthcare: a study for interpreting diabetes prediction, in: R. Misra, R.K. Shyamasundar, A. Chaturvedi, R. Omer (Eds.), Machine Learning and Big Data Analytics, Proceedings of International Conference on Machine Learning and Big Data Analytics, vol. 256, 2021, pp. 95–105.

[17] H. Lu, S. Uddin, Explainable stacking-based model for predicting hospital readmission for diabetic patients, Information 13 (9) (2022) 436.

[18] V. Vishwarupe, P.M. Joshi, N. Mathias, S. Maheshwari, S. Mhaisalkar, V. Pawar, Explainable AI and interpretable machine learning: a case study in perspective, Procedia Comput. Sci. 204 (2022) 869–876.

[19] L.P. Joseph, E.A. Joseph, R. Prasad, Explainable diabetes classification using hybrid Bayesian-optimized TabNet architecture, Computer in Biology and Medicine 151 (Part A) (December 2022).

[20] V. Aelgani, S.K. Gupta, V.A. Narayana, Local agnostic interpretable model for diabetes prediction with explanations using XAI, in: K.A. Reddy, et al. (Eds.), Proceedings of Fourth International Conference on Computer and Communication Technologies, Lecture Notes in Networks and Systems, vol. 606, Springer, Singapore, 2023, pp. 417–425, https://doi.org/10.1007/978-981-19-8563-8_40.

[21 I. Uysal, Interpretable diabetes prediction using XAI in healthcare application, J. Multidiscip. Dev. 8 (1) (2023) 20–38.

[22] S.Y. Lee, W.C.C. Chu, Y.H. Tseng, H.L. Tsai, Explainable AI applied in healthcare: a case study of diabetes prediction, in: IEEE 24th International Conference on Software Quality, Reliability, and Security Companion (QRS-C), 2024.

[23] T.Q. Vinh, H. Byeon, Towards transparent diabetes prediction: unveiling the factors with explainable AI, Int. J. Eng. Trends Technol. 72 (Issue 5) (2024) 26–35, https://doi.org/10.14445/22315381/IJETT-V72I5P103.

[24] Sharia Arfin Tanim, Rafi Aurnob Al, Tahmid Enam Shrestha, MD Rokon Islam Emon, M.F. Mridha, Md Saef Ullah Miah, Explainable deep learning for diabetes diagnosis with DeepNetX2, Biomed. Signal Process Control 99 (2025) 106902, https://doi.org/10.1016/j.bspc.2024.106902. ISSN 1746-8094.

[25] J.H. Friedman, Multivariate adaptive regression splines, Ann. Stat. 19 (1) (1991) 1–67, https://doi.org/10.1214/aos/1176347963. CiteSeerX 10.1.1.382.970.

[26] M.F. Faruque, Asaduzzaman, I.H. Sarker, Performance analysis of machine learning techniques to predict diabetes mellitus, in: 2019 International Conference on Electrical, Computer and Communication Engineering (ECCE), Cox'sBazar, Bangla-Desh, 2019, pp. 1–4, https://doi.org/10.1109/ECACE.2019.8679365.

[27] S. Shafi, G. A. Ansari, Early Prediction of Diabetes Disease & Classification of Algorithms Using Machine Learning Approach (May 25, 2021). Proceedings of the International Conference on Smart Data Intelligence (ICSMDI 2021),https://doi.org/10.2139/ssrn.3852590.

[28] L. Breiman, Random forests, Mach. Learn. 45 (2001) 5–32.

[29] G.A. Pethunachiyar, Classification of diabetes patients using kernel based support vector machines, in: 2020 International Conference on Computer Communication and Informatics (ICCCI), 2020, pp. 1–4, https://doi.org/10.1109/ICCCI48352.2020.9104185. Coimbatore, India.

[30] Z. Zhang, Introduction to machine learning: k-nearest neighbors, Ann. Transl. Med. 4 (11) (2016) 218, https://doi.org/10.21037/atm.2016.03.37.

[31] J.H. Friedman, Greedy function approximation: a gradient boosting machine, Ann. Stat. (2001) 1189–1232.

[32] M. Jamei, M. Ali, H. Afzaal, M. Karbasi, A. Malik, A.A. Farooque, Z. Haydar, Q. U. Zaman, Accurate monitoring of micronutrients in tilled potato soils of eastern Canada: application of an eXplainable inspired-adaptive boosting framework coupled with SelectKbest, Comput. Electron. Agric. 216 (January 2024).
[33] R.C. Deo, O. Kisi, V.P. Singh, Drought forecasting in eastern Australia using multivariate adaptive regression spline, least square support vector machine and M5Tree model, Atmos. Res. 184 (2017) 149–175.
[34] X. Wen, Y. Xie, L. Jiang, Y. Li, T. Ge, On the interpretability of machine learning methods in crash frequency modeling and crash modification factor development, Accid. Anal. Prev. 168 (2022).
[35] L.L.J. Koppes, J.M. Dekker, H.F.J. Hendriks, L.M. Bouter, R.J. Heine, Moderate alcohol consumption lowers the risk of type 2 diabetes, Diabetes Care 28 (3) (March 2005) 719–725.
[36] C. Knott, S. Bell, A. Britton, Alcohol consumption and the risk of type 2 diabetes: a systematic review and dose-response meta-analysis of more than 1.9 million individuals from 38 observational studies, Diabetes Care 38 (9) (September 2015) 1804–1812.
[37] A.J. de Craen, A.M. Oleksik, A.B. Maier, R.G. Westendorp, Causes of health and disease in old age: new insights from the Leiden Research Program on Ageing, Tijdschr. Gerontol. Geriatr. 40 (2009) 237–243.
[38] J.L. Rodgers, J. Jones, S.I. Bolleddu, S. Vanthenapalli, L.E. Rodgers, K. Shah, K. Karia, S.K. Panguluri, Cardiovascular risks associated with gender and aging, J Cardiovasc Dev Dis 6 (2) (2019 Apr 27) 19, https://doi.org/10.3390/jcdd6020019. PMID: 31035613; PMCID: PMC6616540.
[39] B.H. Chew, S.S. Ghazali, M. Ismail, J. Haniff, M.A. Bujang, Age ≥60 years was an independent risk factor for diabetes-related complications despite good control of cardiovascular risk factors in patients with type 2 diabetes mellitus, Exp. Gerontol. 48 (5) (2013) 485–491.
[40] N. Gray, G. Picone, F. Sloan, A. Yashkin, Relation between BMI and diabetes mellitus and its complications among US older adults, South. Med. J. 108 (1) (January 2015) 29–36.
[41] A.O. Odegaard, W.P. Koh, G. Vazquez, K. Arakawa, H.P. Lee, M.C. Yu, M. A. Pereira, BMI and diabetes risk in Singaporean Chinese, Diabetes Care 32 (6) (2009) 1104–1106.
[42] Welcome to the SHAP documentation — SHAP latest documentation,https://shap.readthedocs.io/en/latest/.
[43] Cheng, et al., Tunneling-induced settlement prediction using the hybrid feature selection method for feature optimization, Transportation Geotechnics 36 (September 2022) 100808.
[44] A. Bijlsma-Rutte, F. Rutters, P.J.M. Elders, S.D.M. Bot, G. Nijpels, Socio-economic status and HbA1c in type 2 diabetes: a systematic review and meta-analysis, Diabetes Metabol. Res. Rev. 34 (6) (September 2018).
[45] A. Kautzky-Willer, M. Leutner, J. Harreiter, Sex differences in type 2 diabetes, Diabetologia, Vo. 66 (2023) 986–1002.